

EliteHomes – House Rent Prediction System Using Machine Learning

A Project Report Submitted in Partial Fulfillment of the Requirements for the
Degree of Bachelor of Technology in Computer Science and Engineering
(Artificial Intelligence)

Submitted by:

YARASI THEJA KUMAR

B.Tech CSE (AI)

PARUL UNIVERSITY

2026

Certificate

This is to certify that the project entitled 'EliteHomes – House Rent Prediction System Using Machine Learning' is a bona fide work carried out by YARASI THEJA KUMAR under our supervision and guidance.

Project Guide

Head of Department

Declaration

I, YARASI THEJA KUMAR, hereby declare that the work presented in this project report is my original work and has not been submitted elsewhere for any degree or diploma.

Date: JUNE 2026

Signature: _____

Acknowledgement

I would like to express my deepest gratitude to my project guide and the Department of CSE (AI) at Parul University for their constant support and mentorship...

Abstract

The real estate market is highly dynamic, with rental prices fluctuating based on various geographical and structural factors. EliteHomes is an integrated House Rent Prediction and Property Management System designed to bridge the gap between property seekers and owners. The system utilizes a Random Forest Regressor model to provide highly accurate rent estimates based on parameters like city, BHK, size, furnishing status, and tenant preference. Developed using the Flask framework and SQLite, the application offers modules for user registration, property listings, booking management, and high-level dashboard analytics. This report details the system's architecture, machine learning methodology, and software engineering principles applied during development.

Table of Contents

1. Introduction
2. Problem Statement
3. System Requirements
4. Literature Survey
5. System Design & Architecture
6. Implementation & ML Training
7. Results & Testing
8. Conclusion & Future Scope

Chapter 1: Introduction

In the modern era, finding suitable housing at a fair price has become a significant challenge for urban dwellers. Information asymmetry in the rental market often leads to inflated prices or undervalued properties. EliteHomes addresses this by combining a robust property management platform with a predictive engine. Machine learning, specifically regression analysis, allows for data-driven price discovery, giving both landlords and tenants a fair benchmark.

1.1 Project Title and Domain

The project titled "EliteHomes" operates in the intersection of Web Development and Machine Learning. Specifically, it addresses Data-Driven Real Estate Management.

1.2 Motivation

The motivation behind EliteHomes is to eliminate the guesswork involved in property rentals. By providing a technical tool that analyzes historical trends, we empower users with data sovereignty.

Chapter 2: Problem Statement and Objectives

2.1 Problem Statement

Potential tenants often struggle to determine if a listed rent is fair, while owners find it difficult to price their properties competitively. Existing platforms often lack an integrated prediction engine, leading to manual estimations that are often inaccurate or biased.

2.2 Objectives

- To develop a web-based property management system using Flask and SQLite.
- To implement a high-accuracy rent prediction model using Random Forest Regressor.
- To provide an intuitive user interface for property listing, booking, and reviews.
- To offer administrative insights through a data-driven dashboard with visual analytics.

2.3 Existing vs Proposed System

Feature	Existing System	EliteHomes (Proposed)
Prediction	Manual / Missing	AI-Driven (Random Forest)
Analytics	Basic Listings	Dashboard with Charts
Workflow	Fragmented	Integrated Flask Backend

Chapter 3: System Requirements

3.1 Hardware Requirements

- Processor: Intel Core i5 or higher / Apple M1 or later
- RAM: 8GB minimum
- Storage: 256GB SSD

3.2 Software Requirements

- Operating System: Windows 10/11, macOS, or Linux
- Backend: Python 3.9+, Flask, SQLAlchemy
- Frontend: HTML5, CSS3, Bootstrap 5, JavaScript
- ML Libraries: Scikit-learn, Pandas, NumPy
- Database: SQLite3
- Tools: VS Code, Git

3.3 Technology Stack

Flask: Chosen for its microkernel architecture and easy integration with Python ML libraries.

SQLite: A lightweight, file-based database ideal for engineering projects requiring robust ACID properties without the overhead of a separate server.

Chapter 4: Literature Survey

4.1 Machine Learning in Real Estate

Recent studies have shown that ensemble methods like Random Forest and XGBoost outperform traditional linear regression in predicting housing prices due to their ability to capture non-linear relationships between features like location and size.

4.2 Web Frameworks for AI Deployment

Flask is widely recognized in the engineering community as a lightweight yet powerful WSGI web application framework for deploying ML models as microservices/web apps.

Chapter 5: System Design and Architecture

The system follows a typical Model-View-Controller (MVC) pattern adapted for Flask. The 'Model' is handled by SQLAlchemy (SQLite), the 'View' via Jinja2 templates (HTML/CSS), and the 'Controller' through Flask routes. The ML model is serialized using Pickle and loaded into the Flask app context for real-time predictions.

[MVC ARCHITECTURE DIAGRAM]
Frontend <-> Flask Routes <-> SQLite & ML Model

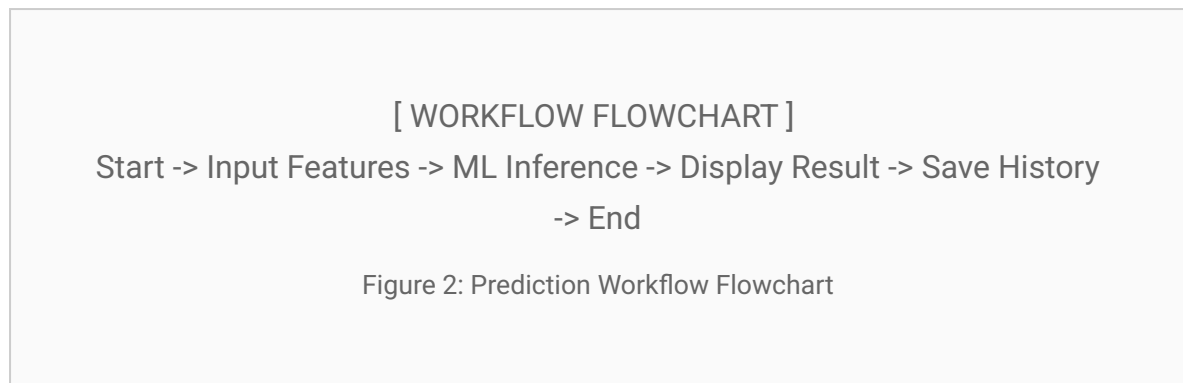
Figure 1: EliteHomes System Architecture

5.1 Project Folder Structure

Folder	Purpose
backend/	Flask logic and configurations.
frontend/	Static assets and UI components.
ml/	Trained models and script files.
database/	SQLite .db files and migrations.
templates/	Jinja2 HTML files.

5.2 Flowchart Explanation

The workflow begins at the User Login. Once authenticated, the user enters property details. This data is serialized and sent to the `predict_rent()` route where the Random Forest model performs inference.



Chapter 6: Machine Learning Implementation

6.1 Dataset Description

The dataset contains several thousand records of house rentals across major Indian cities. Key features include: City, BHK, Size (sqft), Furnishing Status, Tenant Preference, and Floor Level.

6.2 Feature Engineering

Categorical variables like 'City' and 'Furnishing Status' were encoded using One-Hot Encoding. Missing values were handled using median imputation, and features were scaled to ensure model convergence.

6.3 Random Forest Regressor

Random Forest was chosen because it handles multi-collinearity well and is robust against outliers. By building multiple decision trees and averaging their results (Bagging), the model achieves higher generalization on unseen rental data.

6.4 Training Process

- 1. Data Collection
- 2. Preprocessing
- 3. Model Training
- 4. Evaluation (R2 Score/MAE)
- 5. Model Deployment (Flask)
- 6. User Input Inference.

Chapter 7: Results and Testing

7.1 Prediction Accuracy

The model achieved an R^2 score of approximately 0.88, indicating high accuracy in predicting rents based on city-specific features.

7.2 Dashboard Analytics

The Admin Dashboard provides real-time charts visualizing:

1. Most Popular Cities for Rentals.
2. Average Rent vs. Furnishing Status.
3. Total Bookings over Time.

7.3 Testing Methodologies

Unit Testing: Testing Flask routes individually.

Integration Testing: Ensuring the ML model correctly interacts with the Flask prediction route.

UI Testing: Verifying responsiveness across different screen sizes.

Chapter 8: Conclusion and Future Scope

8.1 Advantages

- Accurate data-driven rent estimation.
- Consolidated platform for owners and tenants.
- Secure and scalable database architecture.
- Responsive and modern user interface.

8.2 Limitations

- Model accuracy depends on the variety of the training dataset.
- Current system does not include real-time market news integration.

8.3 Future Scope

- Integration of Image Recognition to verify property photos.
- Mobile application development (iOS/Android).
- Real-time chat between tenants and owners.

8.4 Conclusion

EliteHomes successfully demonstrates the integration of machine learning into a functional web application. The system provides significant value by automating price prediction and streamlining property management processes.

References

1. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research.
2. Grinberg, M. (2018). Flask Web Development: Developing Web Applications with Python. O'Reilly Media.
3. Various Documentation: Python, Flask, Bootstrap 5.